

ODE $\rightarrow$ Ordinary Differential Equation $\rightarrow \frac{d}{dx}$

PDE $\rightarrow$ partial Differential Equation $\rightarrow \frac{\partial}{\partial x}$

Differential Equation: An equation involving derivatives of one or more dependent of one or variables is called a differential equation.

For example:

$$\frac{d^2y}{dx^2} + xy \left(\frac{dy}{dx} \right)^2 = 0$$

$$\frac{\partial v}{\partial s} + \frac{\partial v}{\partial t} = v$$

Ordinary Differential Equation: A differential equation involving ordinary derivatives of one or more dependent variables with respect to a single independent variable is called an ordinary differential equation.

For Example:

$$\frac{d^3 y}{dx^3} + 5 \frac{dy}{dx^2} - \left(\frac{dy}{dx} \right)^2 = 0$$

$$\frac{d^4 x}{dt^4} + \frac{d^3 x}{dt^3} + 3x = \sin t$$

$\frac{d}{dx} \rightarrow$ ଏହା Ordinary independent variable equation.

$\frac{\partial}{\partial x} \rightarrow$ ଏହା partial

Order and Degree of a Differential Equation :

The Order of a differential equation is the order of the highest derivative in the equation.

The degree of a differential equation is the degree of the derivative of the

highest order in the equation.

For example :

$$\textcircled{i} \left(\frac{d^3 y}{dx^3} \right)^1 - 2 \frac{d^2 y}{dx^2} + \frac{dy}{dx} + 13y = 5 \left(\frac{dy}{dx} \right)$$

সুস্পষ্ট করে [This is a equation of order 3]

Equation i) has order 3 and degree 1

Order → কত
মতবার differentiation করতে
হবে

যেখানে order
সেখানেই degree
আমরা।

$$\textcircled{ii} \left(\frac{dy}{dx} \right)^5 + \left(\frac{d^3 y}{dx^3} \right)^5 - \frac{d^2 y}{dx^2} = 57$$

Equation ii) has order 3 and degree 5

$$\textcircled{iii} \frac{d^4 y}{dx^4} - 5 \frac{d^3 y}{dx^3} + \left(\frac{d^2 y}{dx^2} \right)^5 - \left(\frac{dy}{dx} \right)^7 = 2y$$

Equation iii) has order 4 and degree 1

degree and power same

- * Separation of variables
- * Homogeneous Differential Equation
- * Exact Differential Equation
- * Non-exact

DE $\rightarrow$ Differential Equation

Indefinite integral & constant আছে।

$dx, dy, dm, dt \rightarrow$ এগুলো operator

Operator এর সাথে যাই থাকবে সেই variable

যেকোন differential Equation এর math-এ answer পাওয়ার আগ মুহূর্তে integration করতে হবে।

Separation of variables:

Format: $f(x)dx + g(y)dy = 0$

Solve: $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$

Solution:

Given that,

$$\frac{dy}{dx} = \frac{1+y^2}{1+x^2} \quad \text{--- (1)}$$

separating the variables we get,

$$\frac{1}{1+y^2} dy = \frac{1}{1+x^2} dx$$

Now Integrating,

$$\int \frac{1}{1+y^2} dy = \int \frac{1}{1+x^2} dx$$

$$\Rightarrow \tan^{-1} y = \tan^{-1} x + \tan^{-1} C$$

$$\Rightarrow \tan^{-1} y - \tan^{-1} x = \tan^{-1} C$$

$$\Rightarrow \tan^{-1} \frac{y-x}{1+xy} = \tan^{-1} C$$

$$\Rightarrow \frac{y-x}{1+xy} = c$$

$$\Rightarrow y-x = c(1+xy)$$

$$\Rightarrow y = x + c(1+xy)$$

where c is an arbitrary constant. This is the required solution.

Formula:

$$* \int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \frac{x}{a}$$

$$* \int \frac{dx}{1+x^2} = \tan^{-1} x$$

solve $(x^2-4)y^4 dx - x^3(y^2-3)dy = 0$

Solution:

Given that,

$$(x^2-4)y^4 dx - x^3(y^2-3)dy = 0$$

Separating the variables we get,

$$\frac{x^2-4}{x^3} dx - \frac{y^2-3}{y^4} dy = 0$$

$$\Rightarrow \frac{x^2}{x^3} dx - \frac{4}{x^3} dx - \frac{y^2}{y^4} dy + \frac{3}{y^4} dy = 0$$

$$\Rightarrow \frac{1}{x} dx - \frac{4}{x^3} dx - \frac{1}{y^2} dy + \frac{3}{y^4} dy = 0$$

Now integrating we get,

$$\int \frac{1}{x} dx - \int 4x^{-3} dx - \int y^{-2} dy + \int 3y^{-4} dy = 0$$

$$\Rightarrow \ln x - 4 \frac{x^{-2}}{-2+1} - \frac{y^{-2+1}}{-2+1} + 3 \cdot \frac{y^{-4+1}}{-4+1} = 0$$

$$\Rightarrow \log x - 4 \frac{x^{-2}}{-2} - \frac{y^{-1}}{-1} + 3 \frac{y^{-3}}{-3} = 0 \quad \left| \begin{array}{l} \int x^n dx = \frac{x^{n+1}}{n+1} \\ \int e^x dx = e^x \end{array} \right.$$

$$\Rightarrow \log x + \frac{2}{x^2} + \frac{1}{y} - \frac{1}{y^3} = 0$$

solve: $\frac{dy}{dx} = e^x e^{-y} + x^2 e^{-y}$

Solution:

Given that,

$$\frac{dy}{dx} = e^x e^{-y} + x^2 e^{-y}$$

$$\Rightarrow \frac{dy}{dx} = e^{-y} (e^x + x^2)$$

Separating the variables we get,

$$\frac{dy}{e^{-y}} = (e^x + x^2) dx$$

$$\Rightarrow e^y dy = e^x dx + x^2 dx$$

now integrating we get:

$$\int e^y dy = \int e^x dx + \int x^2 dx$$

$$\Rightarrow e^y = e^x + \frac{x^2+1}{2+1}$$

$$\Rightarrow e^y = e^x + \frac{x^3}{3} + c$$

where c is an arbitrary constant. This is the required solution.

Important:

$$\text{solve : } (e^y + 1) \cos x dx + e^y \sin x dy = 0$$

Solution :

Given that,

$$(e^y + 1) \cos x dx + e^y \sin x dy = 0$$

separating the variables we get,

$$\frac{\cos x}{\sin x} dx + \frac{e^y}{e^y + 1} dy = 0$$

Now integrating we get,

$$\int \frac{\cos x}{\sin x} dx + \int \frac{e^y}{e^y + 1} dy = 0$$

$$\Rightarrow \log \sin x + \log (e^y + 1) = \log c$$

$$\Rightarrow \log \{ \sin x (e^y + 1) \} = \log c$$

$$\Rightarrow \sin x (e^y + 1) = c$$

where c is an arbitrary constant. This is the required solution.

Formula:

$$\frac{d}{dx} \sin x = \cos x$$

$$\frac{d}{dy} (e^y + 1) = e^y$$

$$\int \frac{f'(x)}{f(x)} dx = \log f(x)$$

যদি কোন integration
এ সমাধান দায়। তাহলে
নিচের ফাংশনকে
differentiation করলে যদি
উপরের দাঁ আকস্মিক। তখন
integration এর
answer হিসেবে
log নিচের function
লিখতে হবে।

যাতিলাল ঘোষা math

solve: $\frac{dy}{dx} = (4x + y + 1)^{-1}$

solution:

Given that,

$$\frac{dy}{dx} = (4x + y + 1)^{-1} \quad \text{--- (1)}$$

Let us put,

$$4x + y + 1 = v$$

Differentiating with respect to x , we get,

$$\frac{d}{dx} (4x + y + 1) = \frac{dv}{dx} (v)$$

$$\Rightarrow 4 + \frac{dy}{dx} = \frac{dv}{dx}$$

$$\frac{dy}{dx} = \frac{dv}{dx} - 4$$

putting this value in (1)

$$\frac{dv}{dx} - 4 = v^{-1}$$

$$\Rightarrow \frac{dv}{dx} = v^2 + 4$$

$$\Rightarrow \frac{1}{v^2 + 4} dv = 1 dx$$

Formula

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}$$

Now integrating we get,

$$\int \frac{1}{v^2 + 4} dv = \int 1 dx$$

$$\Rightarrow \frac{1}{2} \tan^{-1} \frac{v}{2} = x + C$$

$$\Rightarrow \frac{1}{2} \tan^{-1} \frac{4x + y + 1}{2} = x + C$$

where C is an arbitrary constant. This is the required solution.

"Homogeneous Differential Equation" সমজাতিক (HDE)

এ equation ৩ সমজাতিক degree সমান তাই Homogeneous Differential equation.

1 HDE : An equation of the form $\frac{dy}{dx} = \frac{f_1(x,y)}{f_2(x,y)}$ is called a homogeneous differential equation in which $f_1(x,y)$ and $f_2(x,y)$ are homogeneous functions of x and y of the same degree which can be reduced to an equation in which variables are separable by using $y = vx$, $\frac{dy}{dx} = v + x \frac{dv}{dx}$

$$\frac{xy \, dx}{2} + \frac{yz \, dy}{2} + \frac{xz \, dz}{2} = \frac{m^2 \, dm}{2} \quad \left\{ \text{HDE} \right.$$

$$\frac{x^2 + y^2}{2z^3} \quad \left\{ \begin{array}{l} \text{প্রতিটি term এর Degree সমান} \\ \text{তাই HDE} \end{array} \right.$$

$$y = vx$$

$$\frac{d}{dx} y = \frac{d}{dx} vx$$

$$\Rightarrow \frac{dy}{dx} = v \frac{d}{dx} x + x \frac{dv}{dx}$$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

✓

বামপাশে $\frac{dy}{dx}$

থাকবে

আর যদি বা থাকে

তাহলে কোনো

formula use করে

বামপাশে $\frac{dy}{dx}$ নিজে

আসতে হবে।

Homogeneous & separation এ নিয়ে যেতে হবে।

$$\frac{(x^2 + y^2) dx}{2} = \frac{xy dy}{2} \Rightarrow$$

$$\frac{x^2 + y^2}{xy} = \frac{dy}{dx}$$



এই form এ থাকলে
আর separation করে নিতে
হবে। আর যদি separation
অবস্থায় থাকে তাহলে বামপাশে
 $\frac{dy}{dx}$ কে এটা math এর
solution করতে হবে



এই formula এ থাকলে
তাহলে ready কয়ই আসে
সুস্থ solution করতে হবে
math করার সময় $\frac{dy}{dx}$
কে বামপাশে আনতে
হবে।

Solve: $y^r + x^r \frac{dy}{dx} = xy \frac{dy}{dx}$

Solution:

Given that,

$$y^r + x^r \frac{dy}{dx} = xy \frac{dy}{dx}$$

$$\begin{aligned} y &= vx \\ &= xvx \\ &= vx^2 \end{aligned}$$

$$\Rightarrow y^r = (xy - x^r) \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y^r}{xy - x^r} \quad \text{--- (1)}$$

This is a homogeneous differential equation of degree 2. Let $y = vx$ be the solution of (1). So that $\frac{dy}{dx} = v + x \frac{dv}{dx}$ and from (1) we get,

$$\Rightarrow v + x \frac{dv}{dx} = \frac{v^2 x^r}{vx^r - x^r}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{v^2 x^r}{x^r (v - 1)}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{v^2}{v - 1}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v^2}{v-1} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v^2 - v + v}{v-1}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v}{v-1}$$

$\ln e$ বা $\log e$
 এক অপরের inverse
 তাই একসাথে
 নিশ্চল হয়ে $\emptyset$
 ফোটে যাবে।

Now separating the variables

$$\frac{v-1}{v} dv = \frac{1}{x} dx$$

$$\Rightarrow \frac{v}{v} dv - \frac{1}{v} dv = \frac{1}{x} dx$$

$$\Rightarrow 1 dv - \frac{1}{v} dv = \frac{1}{x} dx$$

Now, Integrating we get

$$\int 1 dv - \int \frac{1}{v} dv = \int \frac{1}{x} dx$$

$$\Rightarrow v - \log v = \log x + \log e$$

$$\Rightarrow \log e^v - \log v = \log x + \log C_1$$

$$\Rightarrow \log \frac{e^v}{v} = \log C_1 x$$

$$\Rightarrow \frac{e^v}{v} = C_1 x$$

$$\Rightarrow \frac{e^{y/x}}{y/x} = C_1 x$$

$$\Rightarrow e^{y/x} = C_1 x \cdot \frac{y}{x}$$

$$\Rightarrow e^{y/x} = C_1 y$$

$$\Rightarrow y = C e^{y/x}$$

$$y = vx$$

$$v = \frac{y}{x}$$

Answer লিখার সময়
সবসময় single y কে
বাক্স পাঠো আনতে হবে

$$y = \frac{1}{C_1} e^{y/x}$$

$$2 = \frac{1}{2} = C$$

where C is an arbitrary constant. This is the required solution.

একোমাত্রার পদের degree টা একসাথে।

Example:

$$(x+y)^2 = 1+1$$

$$x^2 + 2xy + y^2 \quad \left\{ \text{Homogeneous} \right.$$

$x+y \rightarrow$ এই দুইটারই হলো আলাদা আলাদা term
এদের degree হচ্ছে 1.

যদি কোন power বা উৎসের degree বের করতে বলা
তাহলে প্রথম অবস্থানটি দেখাতে হবে। একোমাত্রার
পদের যে সংখ্যাটা আসবে। সেটাটাই আমরা
degree হিসেবে count করবো।

solve: $(x+y) dy + (x-y) dx = 0$

Solution:

Given that,

$$(x+y) dy + (x-y) dx = 0$$

$$\Rightarrow (x+y) dy = -(x-y) dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{y-x}{x+y} \quad \text{--- (1)}$$

This is a homogeneous differential equation of degree 1. Let $y = vx$ be the solution of (1) so that $\frac{dy}{dx} = v + x \frac{dv}{dx}$ and from (1) we get

$$v + x \frac{dv}{dx} = \frac{vx - x}{x + vx}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{x(v-1)}{x(v+1)}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{v-1}{v+1}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v-1}{v+1} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v-1-v(v+1)}{v+1}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v-1-v^v-v}{v+1}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{-(v^v+1)}{v+1}$$

Separating the variables, we get

$$\frac{v+1}{v^v+1} dv = -\frac{1}{x} dx$$

$$\Rightarrow \frac{v+1}{v^v+1} dv + \frac{1}{x} dx = 0$$

$$\Rightarrow \frac{v}{v^v+1} dv + \frac{1}{v^v+1} dv + \frac{1}{x} dx = 0$$

$$\Rightarrow \frac{1}{2} \cdot \frac{2v}{v^v+1} dv + \frac{1}{v^v+1} dv + \frac{1}{x} dx = 0$$

Now Integrating we get,

$$\frac{1}{2} \int \frac{2v}{v^v+1} dv + \int \frac{1}{v^v+1} dv + \int \frac{1}{x} dx = 0$$

$$\Rightarrow \frac{1}{2} \log(v^v+1) + \tan^{-1} v + \log x = c$$

$$\Rightarrow \frac{1}{2} \log\left(\frac{x^v+y^v}{x^v}\right) + \tan^{-1} \frac{y}{x} + \log x = c$$

where c is an arbitrary constant. This is the required solution.

solve: $(x^r + y^r) dx - 2xy dy = 0$

solution:

Given that,

$$(x^r + y^r) dx - 2xy dy = 0$$

$$\Rightarrow 2xy dy = (x^r + y^r) dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^r + y^r}{2xy} \quad \text{--- (1)}$$

This is a homogeneous differential equation of degree 2. Let $y = vx$ be the solution of (1) so that $\frac{dy}{dx} = v + x \frac{dv}{dx}$ and from (1) we get

$$v + x \frac{dv}{dx} = \frac{x^r + v^r x^r}{2x \cdot vx}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{x^r (v^r + 1)}{2vx^r}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{v^r + 1}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v^r + 1}{2v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v^r + 1 - 2v^r}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1-v^2}{2v}$$

Separating the variables we get,

$$\frac{2v}{1-v^2} dv = \frac{1}{x} dx$$

$$\Rightarrow - \frac{(-2v)}{1-v^2} dv = \frac{1}{x} dx$$

$$\Rightarrow \frac{1}{x} dx + \frac{(-2v)}{1-v^2} dv = 0$$

Now, Integrating we get

$$\int \frac{1}{x} dx + \int \frac{(-2v)}{1-v^2} dv = 0$$

$$\Rightarrow \log x + \log(1-v^2) = \log c$$

$$\Rightarrow \log x \cdot (1-v^2) = \log c$$

$$\Rightarrow x(1-v^2) = c$$

$$\Rightarrow x \left(1 - \frac{y^2}{x^2}\right) = c$$

$$\Rightarrow x \left(\frac{x^2 - y^2}{x^2}\right) = c$$

$$\Rightarrow \frac{x^r - y^r}{x} = c$$

$$\Rightarrow x^r - y^r = cx$$

where c is an arbitrary constant. This is the required solution.

Home work:

solve: $(2xy + 3y^r) dx - (2xy + x^r) dy = 0$

Solution:

Given that,

$$(2xy + 3y^r) dx - (2xy + x^r) dy = 0$$

$$\Rightarrow (2xy + 3y^r) dx = (2xy + x^r) dy$$

$$\Rightarrow \frac{dy}{dx} = \frac{2xy + 3y^r}{2xy + x^r} \quad \text{--- (1)}$$

This is homogeneous differential equation of degree 2.

Let $y = vx$ be the solution of (1) so that

$\frac{dy}{dx} = v + x \frac{dv}{dx}$ and from (1) we get,

$$v + x \frac{dv}{dx} = \frac{2x \cdot vx + 3 \cdot vx^2}{2vx + x^2}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{2vx^2 + 3vx^2}{2vx + x^2}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{x(2v + 3x)}{x^2(2v + 1)}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{2v + 3x}{2v + 1}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{2v + 3x}{2v + 1} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{2v + 3x - v(2v + 1)}{2v + 1}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{2v + 3x - 2v^2 - v}{2v + 1}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v + vx}{2v + 1}$$

$$\Rightarrow x(2v + 1)dv = (v + vx)dx$$

$$\Rightarrow \left(\frac{2v + 1}{v + vx} \right) dv = \frac{dx}{x}$$

Now integrating we get,

$$\int \left(\frac{2v + 1}{v + vx} \right) dv = \int \frac{1}{x} dx$$

$$\Rightarrow \log(v + vx) = \log x + \log c$$

$$\Rightarrow \log(v + vx) = \log x = \log c$$

$$\Rightarrow \frac{v + vx}{x} = c$$

$$\Rightarrow \frac{y}{x} + \left(\frac{y}{x} \right)^2 = c$$

$$\Rightarrow \frac{xy + y^2}{x^2} = cx$$

$$\Rightarrow xy + y^2 = cx^3$$

$$\Rightarrow y + xy - cx^3 = 0$$

Here, c is an arbitrary constant. This is the required solution.

Answer

* solve: $(y + \sqrt{x^2 + y^2}) dx - x dy = 0$

Solution:

$$(y + \sqrt{x^2 + y^2}) dx - x dy = 0$$

$$\Rightarrow x dy = (y + \sqrt{x^2 + y^2}) dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{y + \sqrt{x^2 + y^2}}{x} \quad \text{--- (1)}$$

This is a homogeneous differential equation of degree 1. Let $y = vx$ be the solution of (1). so that $\frac{dy}{dx} = v + x \frac{dv}{dx}$ and from (1) we get,

$$v + x \frac{dv}{dx} = \frac{y + \sqrt{x^2 + y^2}}{x}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{vx + \sqrt{x^2 + v^2 x^2}}{x}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{vx + \sqrt{x^2(1 + v^2)}}{x}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{vx + x \sqrt{1 + v^2}}{x}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{x(v + \sqrt{1 + v^2})}{x}$$

$$\Rightarrow v + x \frac{dv}{dx} = v + \sqrt{1 + v^2}$$

$$\Rightarrow x \frac{dv}{dx} = v + \sqrt{1 + v^2} - v$$

$$\Rightarrow x \frac{dv}{dx} = \sqrt{1 + v^2}$$

$$\Rightarrow \frac{dv}{\sqrt{1 + v^2}} = \frac{dx}{x}$$

Integrating with both side and we get,

$$\int \frac{1}{\sqrt{1 + v^2}} dv = \int \frac{1}{x} dx$$

$$\Rightarrow \log(v + \sqrt{v^2 + 1}) = \log x + \log c$$

$$\Rightarrow \log(v + \sqrt{v^2 + 1}) = \log cx$$

$$\Rightarrow v + \sqrt{v^2 + 1} = cx$$

$$\Rightarrow \frac{y}{x} + \sqrt{\frac{y^2}{x^2} + 1} = cx$$

$$\Rightarrow \frac{y}{x} + \sqrt{\frac{y^2 + x^2}{x^2}} = cx$$

$$\Rightarrow \frac{y}{x} + \frac{\sqrt{y^2 + x^2}}{x} = cx$$

$$\Rightarrow \frac{y + \sqrt{y^2 + x^2}}{x} = cx$$

$$\Rightarrow y + \sqrt{y^2 + x^2} = cx^2$$

$$\left| \begin{array}{l} y = vx \\ v = \frac{y}{x} \end{array} \right.$$

$$\therefore y + \sqrt{y^2 + x^2} = cx^2$$

where c is an arbitrary constant. This is the required solution.

Home work

$$\textcircled{1} \text{ solve: } (x^3 + y^2 \sqrt{x^2 + y^2}) dx - \{xy(\sqrt{x^2 + y^2})\} dy = 0$$

degree = 3

Given that,

$$x^3 + y^2 \sqrt{x^2 + y^2} dx - \{xy(\sqrt{x^2 + y^2})\} dy = 0$$

$$\Rightarrow x^3 + y^2 \sqrt{x^2 + y^2} dx = \{xy \sqrt{x^2 + y^2}\} dy$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^3 + y^2 \sqrt{x^2 + y^2}}{xy \sqrt{x^2 + y^2}} \quad \text{--- (1)}$$

Here,

This is a homogeneous differential equation of degree 3. Let $y = vx$ be the solution of $\textcircled{1}$ so that $\frac{dy}{dx} = v + x \frac{dv}{dx}$ and form the equation $\textcircled{1}$ and we get;

Important topic:

Exact differential Equation:

Definition: The differential Equation of the form $Mdx + Ndy = 0$, where both M and N are functions of x and y is said to be an exact differential equation if the condition

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \text{ is satisfied.}$$

$\frac{\partial}{\partial x} \rightarrow$ partial differential equation (ordinary)

$$\begin{aligned} \frac{d}{dx}(x^2 y) &= x^2 \frac{d}{dx} y + y \frac{d}{dx}(x^2) \Rightarrow \text{ordinary} \\ &= x^2 \frac{dy}{dx} + 2xy \end{aligned}$$

$$\frac{\partial}{\partial x}(x^2 y)$$

$$= 2xy$$

$$\frac{d}{dx}(x^2 y - z^2)$$

$$= x^2 \frac{d}{dx} y + y \frac{d}{dx} x^2 - \frac{d}{dx}(z^2)$$

$$= x^2 \frac{dy}{dx} + 2xy - 2z \frac{dz}{dx}$$

Integrating

$$\int \frac{dx}{x} = \int (v \sqrt{1+v^2}) dv$$

$$\Rightarrow \int \frac{1}{x} dx = \int v \cdot (1+v^2)^{\frac{1}{2}} dv$$

$$\Rightarrow \int \frac{1}{x} dx = \int \frac{1}{2} dt \cdot t^{\frac{1}{2}}$$

$$\Rightarrow \int \frac{1}{x} dx = \frac{1}{2} \int t^{\frac{1}{2}} dt$$

$$\Rightarrow \log x + \log c = \frac{1}{2} \cdot \frac{t^{\frac{3}{2}}}{\frac{3}{2}}$$

$$\Rightarrow \log x + \log c = \frac{1}{2} \times \frac{2}{3} \times t^{\frac{3}{2}}$$

$$\Rightarrow \log x + \log c = \frac{1}{3} t^{\frac{3}{2}}$$

$$\Rightarrow \log c x = \frac{1}{3} (v^2 + 1)^{\frac{3}{2}}$$

$$\Rightarrow \log c x = \frac{1}{3} \left(\frac{y^2}{x^2} + 1 \right)^{\frac{3}{2}}$$

$$\Rightarrow 3 \log c x = \left(\frac{y^2 + x^2}{x^2} \right)^{\frac{3}{2}}$$

$$\Rightarrow 3 \log c x = \frac{(y^2 + x^2)^{\frac{3}{2}}}{x^{2 \times \frac{3}{2}}}$$

$$\Rightarrow 3 \log c x = \frac{(y^2 + x^2)^{\frac{3}{2}}}{x^3}$$

$$\Rightarrow 3x^3 \log c x = (x^2 + y^2)^{\frac{3}{2}}$$

here C is an arbitrary constant. This is the required solution.

Let

$$v^2 + 1 = t$$

$$\Rightarrow 2v \cdot \frac{dv}{dt} = 1$$

$$\Rightarrow 2v dv = dt$$

$$\Rightarrow v dv = \frac{1}{2} dt$$

$$\int \frac{x^n}{x^{n+1}} = \frac{x^n}{n+1}$$

$$[\because v = \frac{y}{x}]$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{x^3 + v^2 x^2 \sqrt{x^2 + v^2 x^2}}{x^2 v \cdot x \sqrt{x^2 + v^2 x^2}}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{x^3 + v^2 x^2 \sqrt{x^2 (1 + v^2)}}{x^2 v \sqrt{x^2 (1 + v^2)}}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{x^3 + v^2 x^2 \cdot x \sqrt{1 + v^2}}{x^2 v \cdot x \sqrt{(1 + v^2)}}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{x^3 + v^2 x^3 \sqrt{1 + v^2}}{x^3 v \sqrt{1 + v^2}}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{x^3 (1 + v^2 \sqrt{1 + v^2})}{x^3 v (\sqrt{1 + v^2})}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{1 + v^2 (\sqrt{1 + v^2})}{v (\sqrt{1 + v^2})}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 + v^2 (\sqrt{1 + v^2})}{v (\sqrt{1 + v^2})} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 + v^2 (\sqrt{1 + v^2}) - v^2 \sqrt{1 + v^2}}{v (\sqrt{1 + v^2})}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1}{v \sqrt{1 + v^2}}$$

$$\Rightarrow \frac{dx}{x} = (v \sqrt{1 + v^2}) dv$$

Exact or formate

$$Mdx - Ndy = 0$$

$$x^2 dy - xy^2 dx = 0$$

$$N = x^2$$

$$M = -xy^2$$

dx এর সাথে সংশ্লিষ্ট
Capital M আর dy এর
সাথে Capital N

$$\frac{\partial}{\partial y} M = \frac{\partial}{\partial x} N$$

দুইটি মান সমান হয়ে যায় অতএব exact.

$$Mdx + Ndy = 0$$

$$\frac{\partial M}{\partial y} + \frac{\partial N}{\partial x}$$

* মেগন প্রবর্তি math কে দেখে আমরা কিভাবে বুঝবো যে কোন format এ বসতে হবে?

Ans: Define an integrating Factor.

$$\text{solve} = (5xy + 4y^2 + 1)dx + (x^2 + 2xy)dy = 0$$

এই math কে কোন integrating non exact method দিয়ে সমাধান হবে। যেহেতু অন্য math এ homogeneous বা separate লিখা যায়।

কিন্তু লিখা থাকলেও এই নিয়ম যে হবে তা কিছু নয় আমাদের আস্তে check করতে হবে।

$$\text{solve: } (3x^2 + 4xy)dx + (2x^2 + 2y)dy = 0$$

Solution: Given that,

$$(3x^2 + 4xy)dx + (2x^2 + 2y)dy = 0 \quad \text{--- (1)}$$

Comparing (1) with $Mdx + Ndy = 0$ we get

$$M = 3x^2 + 4xy$$

$$N = 2x^2 + 2y$$

$$\left| \frac{d}{dx}(2xy) = 2y \right|$$

Now,

$$\frac{\partial M}{\partial y} = 4x$$

$$\frac{\partial N}{\partial x} = 4x$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, hence equation (1) is exact.

Equation (1) can be written as,

$$3x^2 dx + 4xy dx + 2x^2 dy + 2y dy = 0$$

$$\text{Rough } \left(\frac{d}{dx}(x^3) = 3x^2 \quad \frac{d}{dx}(2xy) = 4x \right) \left(\frac{d}{dy}(x^2) = 2x \quad \frac{d}{dy}(y^2) = 2y \right)$$

এখানে যেহেতু x এর operator

গঠিত x এর function বসে

differentiation করতে হবে। আর function এ

এখানে যেহেতু y এর operator

গঠিত y এর function বসে differentiation

করতে হবে।

By grouping method we get

$$d(2xy) + 3x^2 dx + 2y dy = 0$$

Now integrating we get,

$$\int d(2xy) + \int 3x^2 dx + \int 2y dy = 0$$

$$\Rightarrow 2xy + 3 \cdot \frac{x^{2+1}}{2+1} + 2 \cdot \frac{y^{2+1}}{2+1}$$

$$\Rightarrow 2xy + 3 \cdot \frac{x^3}{3} + 2 \cdot \frac{y^2}{2}$$

$$= 2xy + x^3 + y^2 = C$$

where C is an arbitrary constant.
This is the required solution.

Rough

Exact এর Format অনুযায়ী math বসান সময়
 $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ মান দুইটা সমান হলেই exact।
তাই

এক operator কে x এর ^{variable} function এর সাথে
differentiation করতে হবে। y এর ^{variable} function এর
সাথে যদি y এর variable থাকে। তাহলে y আপনাপন
নি বসে থাকবে।

① dy অপারেটর কে y এর variable এর সাথে
differentiation করতে হবে। y এর x x যদি
 y এর variable থাকে তাহলে x আপনাপন
বসে থাকবে।

② x এবং y এর আপেক্ষে differentiation বসানো।
যদি যে দুইটার differentiation এর মান বিষয়ে
হয় তাহলে $\Rightarrow d(\text{মান})$ - এটাই grouping method

③ আর যদি যদি মানগুলো না মিলে। তাহলে main
math এর যে মান আছে। সেটাই বসাত হবে grouping
method এর।

④ এরপর integration করে math এর answer
বের করতে হবে। Group বসে function হলে তার integra-
tion বসে লাগবে না। direct শুই মানটাই বসাবে।
Example:

Rough

❶ $3x^3 dx + 4xy dx + 2x^2 dy + 2y^2 dy = 0$
এই math কে আমরা dx এবং dy এর operator
হিসেবে লিখতে পারি -

❶ $3x^3 dx + 4xy dx \rightarrow dx$ এর অপারেটর

❷ $2x^2 dy + 2y^2 dy \rightarrow dy$ এর operator

এখন প্রত্যেক variable কে নিয়ে

❶ dx এর সাপেক্ষে differentiation করতে হবে
যাতে $3x^3$ আসে বা $4xy$ আসে

$$\frac{d}{dx}(x^3) = 3x^2 \quad \frac{d}{dx}(4xy) = 4x$$

❷ $\frac{d}{dy}(2x^2 dy + 2y^2 dy)$ এখানে y এর variable নিয়ে
কিন্তু $2x^2$ আসে 1 আসে করে নিতে হবে

$$\frac{d}{dy}(y) = 1 \quad \frac{d}{dy}(2y^2) = 2y$$

$$3x^3 dx + 2x^2 dy + 4xy dx + 2y^2 dy$$

❸ Grouping method,

$$d(2x^2 y) + 3x^3 dx + 2y^2 dy$$

❹ এরপর integration করলেই math
complete.

❺ $x^3 dx, y^2 dy$ এর ক্ষেত্রে
grouping হচ্ছে না তাই
মানি math এর ক্ষেত্রে
টাই করতে হবে।

$d(2x^2 y) \rightarrow$ এখানে d টি হচ্ছে ordinary,
এবং y এর সাপেক্ষে integration হচ্ছে।

Solve: $(y^4 + 4x^3y + 3x)dx + (x^4 + 4xy^3 + y + 1)dy = 0$

Solution:

Given that,

$$(y^4 + 4x^3y + 3x)dx + (x^4 + 4xy^3 + y + 1)dy = 0 \quad \text{--- (1)}$$

Comparing (1) with $Mdx + Ndy = 0$ we get

$$M = y^4 + 4x^3y + 3x$$

$$N = x^4 + 4xy^3 + y + 1$$

Now,

$$\frac{\partial M}{\partial y} = 4y^3 + 4x^3$$

$$\frac{\partial N}{\partial x} = 4x^3 + 4y^3$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, hence equation (1) is exact.

Equation (1) can be written as,

$$y^4 dx + 4x^3 y dx + 3x dx + x^4 dy + 4xy^3 dy + y dy + 1 dy = 0$$

$$\frac{1}{2}y^5 + \frac{1}{2}x^5 + \frac{3}{2}x^2 + \frac{1}{2}y^2 + y = C$$

By grouping method we get,

$$d(xy^4) + d(x^4y) + 3x dx + y dy + 1 dy$$

Now integrating we get,

$$\int d(xy^4) + \int d(x^4y) + \int 3x dx + \int y dy + \int 1 dy$$

$$= xy^4 + x^4y + \frac{3}{2}x^2 + \frac{1}{2}y^2 + y = C$$

where C is an arbitrary constant

This is the required solution.

Solve $(e^{2x}y^v - 2x)dx + e^{2x}y dy = 0$

Solution:

Given that,

$$(e^{2x}y^v - 2x)dx + e^{2x}y dy = 0 \quad \text{--- (1)}$$

comparing (1) with $mdx + Ndy = 0$ we get,

$$M = e^{2x}y^v - 2x$$

$$N = e^{2x}y$$

Now

$$P \begin{cases} \frac{\partial M}{\partial y} = 2e^{2x}y \\ \frac{\partial N}{\partial x} = 2e^{2x}y \end{cases}$$

$$\frac{d}{dx} e^{2x} = 2e^{2x}$$

$$\frac{d}{dx} e^{2x} = 2e^{2x}$$

$$\frac{d}{dx} (-x^2) = -2x$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ hence equation (1) is exact.

Equation (1) can be written as,

$$e^{2x}y^v dx - 2x dx + e^{2x}y dy = 0$$

By grouping method we get,

$$d\left(\frac{1}{2}e^{2x}y^v\right) - 2x dx = 0$$

Now integrating we get,

$$\int d\left(\frac{1}{2}e^{2x}y^v\right) - \int 2x dx = 0$$

$$\Rightarrow \frac{1}{2}e^{2x}y^v - 2 \cdot \frac{1}{2}x^2 = c$$

$$\Rightarrow \frac{1}{2}e^{2x}y^v - x^2 = c$$

where c is an arbitrary constant. This is the required solution.

Home work:

Lecture: 07

Date: 27.05.23

Solve:

$$(2x \cos y + 3x^2y) dx + (x^3 - x^2 \sin y - y) dy = 0$$

Solution:

$$\frac{d}{dx} x^n = nx^{n-1}$$

$$\frac{d}{dx} x = 1$$

$$\frac{d}{dx} e^{2x} = 2e^{2x}$$

$$\textcircled{10} \frac{d}{dt} \left(\frac{1}{4} t^4 \right) = t^3$$

$$= \frac{1}{4} \cdot 4 t^3$$

$$\frac{d}{dx} \sin x = \cos x \quad \frac{d}{dx} e^{7x} = 7e^{7x}$$

$$\frac{d}{dx} \cos x = -\sin x \quad \left| \begin{array}{l} \frac{d}{dx} \frac{1}{2} e^{2x} \\ = \frac{1}{2} \cdot 2 e^{2x} \\ = e^{2x} \end{array} \right.$$

$$\textcircled{11} \frac{d}{dx} x^{-7} = -7x^{-8}$$

⑩ t^3 এর পাওয়ার positive। power positive হলে differentiation করলে power 1 বাড়িয়ে দিব t^4 দিয়ে differentiation করতে হবে। আর বাকিটা Balance করতে হবে।

⑪ x^{-7} এর পাওয়ার Negative। power Negative হলে differentiation করলে power 1 বামিয়ে দিব x^{-8} দিয়ে differentiation করতে হবে। আর বাকিটা Balance করতে হবে।

$$\Rightarrow 4xy$$

$$\frac{d}{dx} (2x^2y)$$

$$= 2 \cdot 2xy$$

$$= 4xy$$

$$\frac{d}{dx} \left(\frac{3}{2} x^2 \right)$$

$$= \frac{3}{2} \cdot 2x$$

$$= 3x$$

$$\Rightarrow -2x$$

$$\frac{d}{dx} (-x^2)$$

$$= -2x$$

$$\Rightarrow e^{2x}$$

$$\frac{d}{dx} \left(\frac{1}{2} e^{2x} \right)$$

$$= \frac{1}{2} \cdot 2 e^{2x}$$

$$= e^{2x}$$

Solve: $[\cos x + \tan y + \cos(x+y)] dx + [\sin x + \sec y + \cos(x+y)] dy = 0$

Solution:

Given that,

$$[\cos x \tan y + \cos(x+y)] dx + [\sin x + \sec y + \cos(x+y)] dy = 0$$

Comparing (1) with $mdx + Ndy = 0$ we get,

$$m = [\cos x \tan y + \cos(x+y)]$$

$$N = \sin x + \sec y + \cos(x+y)$$

Now,

$$\frac{\partial m}{\partial y} = \cos x \sec^2 y - \sin(x+y)$$

$$\frac{\partial N}{\partial x} = \cos x \cdot \sec y - \sin(x+y)$$

Since, $\frac{\partial m}{\partial y} = \frac{\partial n}{\partial x}$ hence equation (1) is exact.

① is exact.

$$\frac{d}{dx} x^n = nx^{n-1}$$

x^n ତି x ର ଏକ Algebraic function

$\frac{d}{dx}(\cos x) = -\sin x$
 Trigonometric function
 এর সাথে trigonometric f.
 ও আছে। তাই দুই বার differentiation করা যায়।
 প্রথমবার Trigonometric f.
 হলে। দ্বিতীয়বার অল্প Algebraic f.

Equation (1) can be written as,

$$\frac{d}{dx} \{ \cos(x+y) \} = -\sin(x+y) \frac{d}{dx} (x+y)$$

$$\cos x \tan y dx + \cos(x+y) dx + \sin x \sec y dy = -\sin(x+y)$$

$$+ \cos(x+y) dy = 0$$

$$\sin(x+y)$$

By grouping method we get,

$$d(\sin x \tan y) + d\{\sin(x+y)\} = 0$$

now Integrating we get,

$$\int d(\sin x \tan y) + \int d(\sin(x+y)) = 0$$

$$\Rightarrow \sin x \tan y + \sin(x+y) = 0$$

where c is an arbitrary constant. This is the required solution.

$$\# (x^2 + y^2 + x) dx + xy dy = 0$$

Solve:

Given that,

$$(x^2 + y^2 + x) dx + xy dy = 0 \quad \text{--- (1)}$$

comparing (1) with $Mdx + Ndy = 0$ we get,

$$M = x^2 + y^2 + x$$

$$N = xy$$

Now,

$$\frac{\partial M}{\partial y} = 2y, \quad \frac{\partial N}{\partial x} = y$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x} ; \text{ Non exact.}$$

এই math টি solve করার ক্ষেত্রে পেরাম $\frac{\partial M}{\partial y}$ আর $\frac{\partial N}{\partial x}$ এর differentiation মান মিলে যাচ্ছে না। যদি না মিলে তাহলে non-exact.

আর যদি non-exact হয়ে যায়। তবে Integrating factor এর rules follow করার exact বানাতো হবে math টি।

তারপর solve করতে হবে।

Integrating factor এক প্রকার function.

Important Definition:

I.F =

Integrating factor: If an equation becomes exact after it has been multiplied by a function of x and y then such a function is called an integrating factor.

Rules for finding the integrating factors:

$$1. \text{ If } \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = f(x), \text{ A function of } x$$

$$\text{only then, } I.F = e^{\int f(x) dx}$$

$$2. \text{ If } \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{M} = f(y), \text{ A function of } y$$

$$\text{only then } I.F = e^{\int f(y) dy}$$

$$I.F \times (x^2 + y^2 + x) dx + xy dy = 0 \Rightarrow \text{Exact}$$

$$h(x, y, z) = f(x, y, z)$$

variable

$$f(x, y, z) = x + xz - y/z$$

$$g(t) = \sin t \leftarrow \text{function of } x \text{ only}$$

$$g(x) = \cos x \leftarrow \text{function of } x \text{ only}$$

$$f(y) = e^{-y} \leftarrow \text{function of } y \text{ only}$$

$(x^3 + y^3 + x)dx + xydy = 0$ is non-exact
 Exact solution is

Integrating factor rules follow below

$$\textcircled{1} \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{2y - y}{xy} = \frac{y}{xy} = \frac{1}{x} = f(x)$$

only x is function. Hence Rules ① & ② are function rules. Hence ③ Number rules - 4

$$\Rightarrow e^{\int \frac{1}{x} dx} \Rightarrow e^{\log x} = x$$

$$I.F \times (x^3 + y^3 + x)dx + xydy = 0$$

$$\Rightarrow x \times (x^3 + y^3 + x)dx + x \cdot xydy = 0$$

$$\Rightarrow (x^4 + xy^3 + x^2)dx + x^2ydy = 0 \quad \text{--- ②}$$

Rough use of the solution is exact solution is

$$M = x^4 + xy^3 + x^2 \quad \frac{\partial M}{\partial y} = 2xy$$

$$N = x^2y \quad \frac{\partial N}{\partial x} = 2xy$$

And now equation ② is exact from ② we get

$$(x^4 + xy^3 + x^2)dx + x^2ydy = 0$$

$$\Rightarrow x^4 dx + xy^3 dx + x^2 dx + x^2y dy = 0$$

$$\frac{1}{5}x^5 + \frac{1}{2}x^2y^2 + \frac{1}{3}x^3 + \frac{1}{2}x^2y^2 = 0$$

By grouping method we get,

$$d(\frac{1}{2}x^2y^2) + x^3 dx + x^2 dx = 0$$

Integrating method we get,

$$\int (\frac{1}{2}x^2y^2) + \int x^3 dx + \int x^2 dx = 0$$

$$\Rightarrow \frac{1}{2}x^2y^2 + \frac{x^4}{4} + \frac{x^3}{3} = 0$$

where c is an arbitrary constant. This is the required solution.

Non Exact

Solve: $(x^2 + y^2 + x)dx + xydy = 0$

Solution:

Given that,

$$(x^2 + y^2 + x)dx + xydy = 0 \quad \text{--- (1)}$$

comparing (1) with $Mdx + Ndy = 0$ we get,

$$M = x^2 + y^2 + x$$

$$N = xy$$

Now

$$\frac{\partial M}{\partial y} = 2y, \quad \frac{\partial N}{\partial x} = y$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ equation (1) is not exact.

Now,

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{2y - y}{xy} = \frac{y}{xy} = \frac{1}{x}$$

which is a function of x only. Therefore

$$\begin{aligned} \text{Integrating factor} &= e^{\int \frac{1}{x} dx} \\ &= e^{\log x} \\ &= x \end{aligned}$$

Multiplying (1) by Integrating factor the equation becomes,

$$x(x^2 + y^2 + x)dx + x \cdot xy \cdot dy = 0$$

$$\Rightarrow (x^3 + xy^2 + x^2)dx + x^2ydy = 0 \quad \text{--- (2)}$$

And now equation 2 is exact. From 2, we get,

$$\frac{x^3 dx}{\frac{1}{4}x^4} + \frac{xy^2 dx}{\frac{1}{2}x^2y^2} + \frac{x^2 dx}{\frac{1}{3}x^3} + \frac{x^2y dy}{\frac{1}{2}x^2y^2} = 0$$

By grouping method,

$$d\left(\frac{1}{2}xy^2\right) + x^3dx + x^2dx = 0$$

Now integrating we get,

$$\frac{1}{2}xy^2 + \frac{1}{4}x^4 + \frac{1}{3}x^3 = C$$

where C is an arbitrary constant. This is the required solution.

$$\text{Solve: } (2y^2 + 3x)dx + 2xy dy = 0$$

Solution:

Given that,

$$(2y^2 + 3x)dx + 2xy dy = 0 \quad \text{--- ①}$$

Comparing ① with $mdx + Ndy = 0$ we get,

$$m = 2y^2 + 3x$$

$$N = 2xy$$

Now

$$\frac{\partial m}{\partial y} = 4y, \quad \frac{\partial N}{\partial x} = 2y$$

Since, $\frac{\partial m}{\partial y} \neq \frac{\partial N}{\partial x}$ equation ① is not exact.

Now,

$$\frac{\frac{\partial m}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{4y - 2y}{2xy} = \frac{2y}{2xy} = \frac{1}{x}$$

which is a function of x only.

$$\begin{aligned}\text{therefore, integrating factor} &= e^{\int \frac{1}{x} dx} \\ &= e^{\log x} \\ &= x\end{aligned}$$

multiplying ① by integrating factor
the equation becomes,

$$\begin{aligned}x(2y^r + 3x)dx + x \cdot 2xy dy &= 0 \\ \Rightarrow (2xy^r + 3x^r)dx + 2x^2y dy &= 0 \quad \text{--- ②}\end{aligned}$$

And now equation ② is exact from
② we get,

$$2xy^r dx + 3x^r dx + 2x^2y dy = 0$$

By grouping method,

$$d(x^2y^r) + 3x^r dx = 0$$

Now integrating we get,

$$\begin{aligned}\int d(x^2y^r) + \int 3x^r dx &= 0 \\ \Rightarrow x^2y^r + x^3 &= c\end{aligned}$$

where c is an arbitrary constant. This
is the required solution.

g#Solve: $(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0$

Solution:

Given that,

$$(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0 \quad \text{--- ①}$$

Comparing ① with $mdx + ndy = 0$ we get

$$m = y^4 + 2y$$

$$N = xy^3 + 2y^4 - 4x$$

Now,

$$\frac{\partial M}{\partial y} = 4y^3 + 2, \quad \frac{\partial N}{\partial x} = y^3 - 4$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, equation 1 is not exact.

Now,

$$\begin{aligned} \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{M} &= \frac{4y^3 + 2 - y^3 + 4}{y^4 + 2y} = \frac{3y^3 + 6}{y^4 + 2y} \\ &= \frac{3(y^3 + 2)}{y(y^3 + 2)} = \frac{3}{y} \end{aligned}$$

which is a function of y only. Therefore

$$\begin{aligned} \text{I.F} &= e^{\int \frac{3}{y} dy} \\ &= e^{3 \int \frac{1}{y} dy} \\ &= e^{3 \log y} \\ &= e^{\log y^3} \\ &= \frac{1}{y^3} \end{aligned}$$

multiplying ① by I.F the equation becomes

$$\begin{aligned} \frac{1}{y^3} (y^4 + 2y) dx + \frac{1}{y^3} (xy^3 + 2y^4 - 4x) dy &= 0 \\ \left(\frac{y^4}{y^3} + \frac{2y}{y^3} \right) dx + \left(\frac{xy^3}{y^3} + \frac{2y^4}{y^3} - \frac{4x}{y^3} \right) dy &= 0 \\ \left(y + \frac{2}{y^2} \right) dx + \left(x + \frac{2y}{y^2} - \frac{4x}{y^3} \right) dy &= 0 \quad \text{--- ②} \end{aligned}$$

And now equation ② is exact. From ② we get,

$$y dx + 2y^{-2} dx + x dy + 2y dy - 4xy^{-3} dy = 0$$

By grouping method,

$$d(xy) + d(2xy^{-2}) + 2y dy = 0$$

Now integrating we get,

$$\int d(xy) + \int d(2xy^{-2}) + \int 2y dy = 0$$

$$= xy + 2xy^{-2} + y^v = c$$

where c is an arbitrary constant.
This is the required solution.

Home work

Define an integrating Factor.

$$\text{Solve } (5xy + 4y^v + 1)dx + (x^v + 2xy)dy = 0$$

Solution:

Given that,

$$(5xy + 4y^v + 1)dx + (x^v + 2xy)dy = 0 \quad \text{--- (1)}$$

Comparing (1) with $Mdx + Ndy = 0$ we get,

$$M = 5xy + 4y^v + 1$$

$$N = x^v + 2xy$$

Now,

$$\frac{\partial M}{\partial y} = 5x + 8y$$

$$\frac{\partial N}{\partial x} = 2x + 2y$$

Since, $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ equation (1) is not exact.

Now,

$$\begin{aligned} \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} &= \frac{5x + 8y - 2x - 2y}{x^v + 2xy} = \frac{3x + 6y}{x^v + 2xy} \\ &= \frac{3(x + 2y)}{x(x + 2y)} = \frac{3}{x} \end{aligned}$$

which is a function of x only. Therefore,

Integrating factor, I.F. = $e^{\int f(x)dx}$

$$= e^{\int 3 \frac{1}{x} dx}$$

$$= e^{3 \log x}$$

$$= e^{\log x^3}$$

$$= x^3$$

Multiplying ① by I.F the equation becomes

$$x^3(5xy + 4y^2 + 1)dx + x^3(x^2 + 2xy)dy = 0$$

$$\Rightarrow (5x^4y + 4x^3y^2 + x^3)dx + (x^5 + 2x^4y)dy = 0$$

$$\Rightarrow 5x^4y dx + 4x^3y^2 dx + x^3 dx + x^5 dy + 2x^4y dy = 0 \quad \text{--- ②}$$

And now equation ② is exact. From ② we get,

$$\Rightarrow 5x^4y dx + 4x^3y^2 dx + x^3 dx + x^5 dy + 2x^4y dy = 0$$

$$\left[x^5y \quad x^4y^2 \quad \frac{1}{4}x^4 \quad yx^5 \quad y^2x^4 \right] \text{ Rough}$$

By grouping method,

$$d(x^5y) + d(x^4y^2) + x^3 dx = 0$$

Now integrating we get,

$$\int d(x^5y) + \int d(x^4y^2) + \int x^3 dx \\ = x^5y + x^4y^2 + \frac{x^4}{4} = C$$

$$\Rightarrow x^5y + x^4y^2 + \frac{1}{4}x^4 = C$$

where C is an arbitrary constant. This is the required solution.

Date: 29-05-2023
repeat math

$$\text{Solve: } (2y^2 + 3x) dx + 2xy dy = 0$$

Solution:

Given that,

$$(2y^2 + 3x) dx + 2xy dy = 0 \quad \text{--- ①}$$

Comparing ① with $Mdx + Ndy = 0$ we get,

$$M = 2y^2 + 3x$$

$$N = 2xy$$

Now,

$$\frac{\partial M}{\partial y} = 4y$$

$$\frac{\partial N}{\partial x} = 2y$$

Since, $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ the equation ① is not-exact

Now,

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{4y - 2y}{2xy} = \frac{2y}{2xy} = \frac{1}{x}$$

which is a function of x only. Therefore

$$\begin{aligned} I.F. &= e^{\int f(x) dx} \\ &= e^{\int \frac{1}{x} dx} \\ &= e^{\log x} \\ &= x \end{aligned}$$

✱ multiplying (1) by I.F. the equation becomes

$$\begin{aligned} x(2y^r + 3x) dx + x \cdot 2xy dy &= 0 \\ \Rightarrow (2xy^r + 3x^r) dx + 2x^r y dy &= 0 \quad \text{--- (ii)} \end{aligned}$$

$\frac{\partial M}{\partial y} = 4xy$
 $\frac{\partial N}{\partial x} = 4xy$

Now equation (ii) is exact. From (ii) we get

$$\begin{aligned} 2xy^r dx + 3x^r dx + 2x^r y dy &= 0 \\ x^r y^r & \quad x^3 & \quad y^r x^r \end{aligned}$$

By grouping method,

$$d(x^r y^r) + 3x^r dx = 0$$

Now integrating we get,

$$\begin{aligned} \int d(x^r y^r) + \int 3x^r dx &= 0 \\ \Rightarrow x^r y^r + 3 \frac{x^3}{3} &= c \\ \Rightarrow x^r y^r + x^3 &= c \end{aligned}$$

where c is an arbitrary constant. This is the required solution.